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Profitability analysis of power generation using waste heat of sponge iron process

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ABSTRACT

The present paper proposes two energy integration options, Option-1 and Option-2, to compare power generation and preheating scheme in sponge iron process. Option-1 accounts power generation whereas in Option-2 preheating of feed material, slinger coal and kiln air is considered. These options utilize heat of waste gas that exits the after burning chamber of the process, which is available from 994 °C to 250 °C. The process integration principles are applied to conserve energy in Option-1 and Option-2. For Option-1 8.97 MW power is generated using heat of waste gas, which gives payback period as 4.17 years. On the other hand, Option-2 consumes 37.4% and 4.7% less coal and water, respectively, in comparison to the existing system. The payback period for option-2 is 3.22 years. The feasibility analyses of these options are also discussed. Based on payback period preheating is more profitable option in comparison to power generation. However, when net profits of two options are compared, power generation option is more profitable. The results of the present study are compared well with that of published work.

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1. Introduction

Sponge iron is a metallic mass produced through direct reduction of iron ore or pellets in the solid state. Its structure is like honeycomb and thus, it is called 'Sponge Iron'. The quality of sponge iron depends primarily on the percentage of metallization, which is ratio of metallic iron to total iron present in the product [1]. It is used as prime source in steel making process.

It is found in the literature that required amount of energy for rotary kiln is in the range from 14.63 GJ to 20.9 GJ per tonne of sponge iron [2]. Although, during the operation enormous amount of heat is generated, which remains unutilized in the process. Therefore, it is clear that energy problems are most important issues in these industries. Effectively heat should be integrated in various ways for these industries. One way is to utilize heat of waste gas in the process to reduce coal consumption. On the other hand, power can be generated using this heat. The literature related to these is discussed hereunder:

The heat of waste gas can be utilized in various ways in the sponge iron process as suggested by many investigators. Bandyopadhyay et al. [3] found coal as principal raw material for the sponge iron process and if very high amount of volatile matter (VM) coal is used, majority of VM is lost through the kiln mouth with hot waste gases. They revealed that 30–40% of total kiln heat is lost with waste gases. Elsenheimer and Serbent [4] also confirmed this fact and found that the energy contained in the form of sensible or chemical energy in waste gas was up to 40%. They recommended a number of options to recuperate the waste gas energy but did not show its practical complications and how much amount of energy could be saved by installing these options.

Hajidavalloo and Alagheband [5] discussed the thermal analysis of sponge iron through electric arc furnace (EAF) route. A new initiative technique was introduced for improving the performance of EAF where raw materials were preheated using heat of waste gas before entering the furnace. It was found that the energy consumption was reduced and productivity was increased by 14% and 13%, respectively.

An energy survey of a sponge iron plant was carried out by Eriksson and Larsson [6]. They suggested various ways to enhance energy efficiency of the process such as internal use, external energy supply or power generation. They also proposed modification in rotary kiln design to increase efficiency. On the other hand, Mignard and Pritchard [7] analyzed the sponge iron process for transmission and storage of remotely generated marine energy.

A few [8] authors suggested that 10–12% amount of energy could be saved by controlling axial and radial air injection. Further,

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Biswas et al. [8] suggested that air jet seal at specific locations of kiln instead of mechanical seal could be used to reduce power consumption.

Kumar and Khanam [9] suggested a plate heat exchanger for carrying out the preheating of air up to 170 °C using heat of waste gas. Due to this coal consumption and waste gas were reduced by 8.7% and 16.7%, respectively. They also proposed a water bath for the utilization of heat content of the kiln discharge. As a result, heavy evaporation losses were minimized, which showed 96.3% reduction in water consumption as compared to the existing system. Prasad et al. [10] proposed two modifications such as case-1and case-2. Preheating of feed material was accounted in case-1, whereas that of feed material as well as air was considered in case-2. Here, case-2 was selected as the best heat recovery option as compared to case-1 and recovery system. It reduced coal and water consumption by 30.5% and 72.6%, respectively. Prasad et al. [11] identified three feasible cases for energy conservation. The best design incorporated preheating of air as well as feed material to rotary kiln using waste gas. Further, it was used to cool kiln outlet and consequently, 12.5% reduction in coal was observed in comparison to the existing system.

Jena et al. [12] suggested a quantitative analysis considering waste gas composition, air requirement, dust loss, etc. They found that thermal efficiency of the process was 46%. Around 33% of the heat of kiln was lost with the waste gas. The authors suggested that heat associated with waste gas may be utilized for power generation. However, they did not show the computation to meet this fact. Agarwal and Sood [13] found that only 35% heat was used in reduction out of the total heat generated from coal. They installed waste heat recovery boiler (WHRB) for tapping this heat and generating process steam. However, they faced a few problems such as: (i) During start up, generated steam quality is not satisfactory for turbine application, (iii) the accretion formation in the kiln, unavailability of feedstock or shut down disturbs the steam generation, and (iv) kiln waste gas contains 35×10^6 to 40×10^6 kg/m^3 of dust particles. Further, Ulrich and Tondon [2] mentioned that the feasibility for one 500 tpd module plant containing a WHRB would be questionable. It should be practicable for the plants beyond the production capacity of 1000 tpd of sponge iron. In fact, this fact was also supported by Agarwal and Sood [13].

A few studies have been carried out on utilizing WHRB in other plants than sponge iron process. Xiuwen et al. [14] investigated energy and exergy efficiency for the cement production systems. The energy efficiencies of raw material preparation system, pulverized coal preparation system, and rotary system were found as 39.4%, 10.8%, and 50.2% respectively, whereas exergy efficiencies were 4.5%, 1.4% and 33.7%, respectively, before using WHRB. On the other hand, after installing WHRB energy efficiencies of raw material preparation system, pulverized coal preparation system, and rotary system were predicted as 45.8%, 15.5% and 55.1% respectively and exergy efficiencies were 7.8%, 2.8% and 38.1% respectively. It recovered 3.7% of total input energy of a rotary kiln and improved exergy efficiencies of the system. They also identified the energy saving schemes for waste heat power generation project system such as degree of resource, energy utilization and energy saving effect. Yufeng et al. [15] proposed three thermodynamic models such as single pressure, dual pressure and energy integration for optimization of heat recovery using industrial waste gas heat. A hierarchical strategy related to energy integration of total site was proposed for improving power generation along with fuel saving. They observed that energy integration system was more efficient than other two strategies.

These studies utilized the waste heat available in sponge iron process for producing power. However, the authors did not check whether power generation using waste heat is profitable or not. The present paper focuses on this aspect of heat recovery through detailed profitability analysis.

2. Sponge iron process

The schematic of sponge iron process is shown in Fig. 1 where raw material enters to the kiln from feed end and moves toward the discharge end due to rotation and inclination of kiln. Sponge iron exits from discharge end. Air enters the kiln as primary and secondary air, which is due to the process requirement, as shown in Fig. 1. Solid products from the rotary kiln enter into the rotary cooler where temperature of products comes down from 1067 °C to 110 °C in a non-oxidizing atmosphere.

The waste gas flows in the opposite direction of the products, which is maintained by induced draft fan mounted before the chimney. Generally, it comes out of the kiln at 900 °C as shown in Fig. 1 and then goes to after burner chamber (ABC) and a horizontal dust settling chamber (DSC) which is located below to ABC. DSC removes large dust particles by gravity. At the end of DSC, waste gases change their flow direction and move upward into combustion area of ABC. Here, combustibles mixed with excess fresh air and burnt completely and acquired temperature 994 °C.

Water is sprayed in the ABC, for removing toxic components and dust particles. As a result, temperature of waste gas is reduced. For this purpose, eight to ten water guns is fitted at different height. Dust particles settle down due to increase in weight, as pressurized water coming from guns falls on waste gas carrying dust particle.

The evaporating cooler (EC) is connected with ABC as shown in Fig. 1. The waste gas is extinguished with water. Eight to ten water guns are attached at the top position of EC around the circumference at same height. For reducing waste gas temperature water is sprayed at desired levels. The bottom part of ABC, DSC and EC are attached with wet scraper, for collecting dust. Further, waste gas is entered to ESP (electrostatic precipitator) for final cleaning. The waste gas temperature is maintained below 250 °C. The exit of ESP is connected to the chimney through which filtered waste gas is directly released to the atmosphere. The specific heats of the different components and cost data are given in Tables 1 and 2, respectively. The following equations are involved in the rotary kiln are mentioned below:

$$Fe_2O_3 + CO = 2FeO + CO_2 \quad (1)$$

$$FeO + CO = Fe + CO_2 \quad (2)$$

$$2CO + O_2 = 2\,CO_2 \quad (3)$$

$$CO_2 + C \leftrightarrow 2CO \quad (4)$$

$$C + O_2 \leftrightarrow CO_2 \quad (5)$$

$$2C + O_2 = 2CO \quad (6)$$

$$2H_2 + O_2 = 2H_2O \quad (7)$$

3. Heat recovery options

In the present work two options, Option-1 and Option-2, are proposed for energy conservation using heat of waste gas of sponge iron process. In the existing system waste gas decreases its temperature from 994 °C to 250 °C using water in EC as shown through Fig. 1. It carries significant amount of heat, which can be utilized in the process for energy conservation. Option-1 utilizes heat of waste

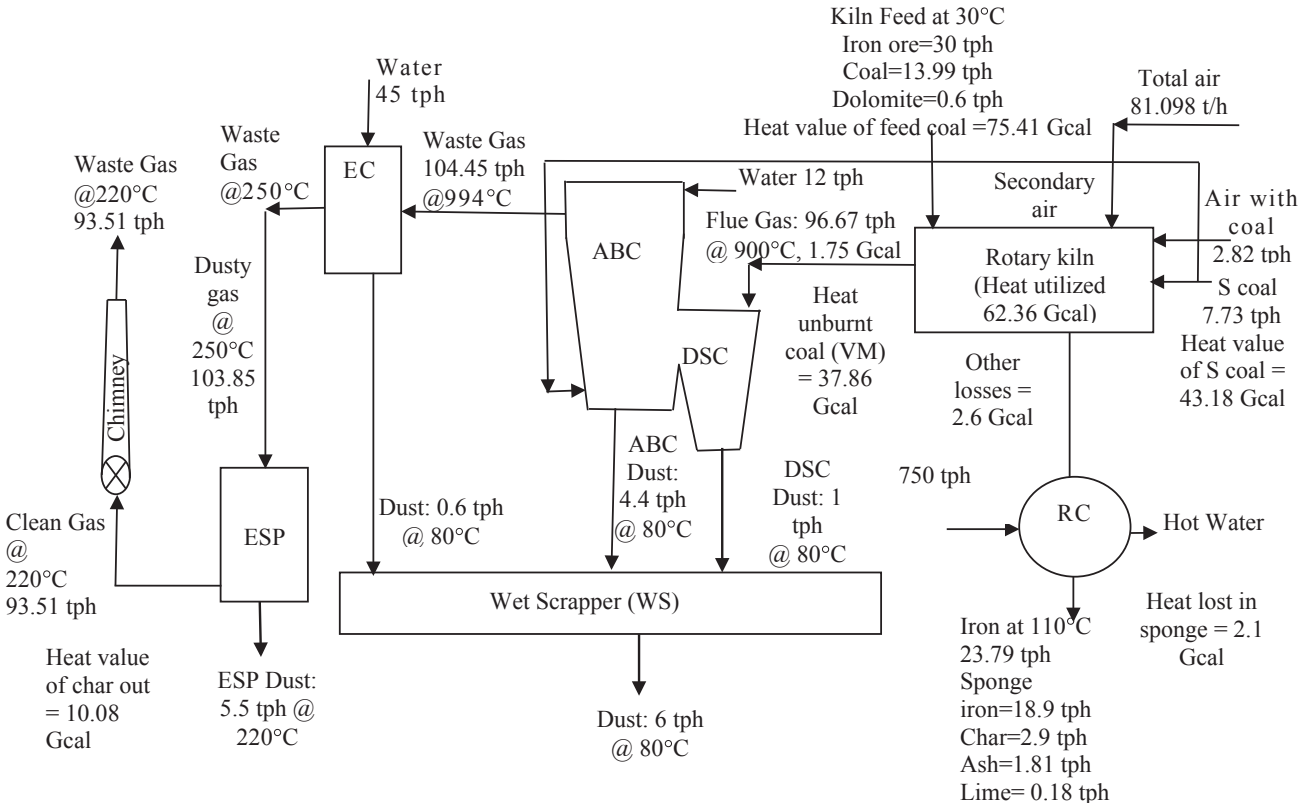


Fig. 1. Process Flow Diagram of the sponge iron process.

Table 1
Specific heat of the different streams.

Stream	Specific heat (J/kg °C)	References
Kiln feed	1092.7	[25–27]
Kiln air	1032.6	[25]
Slinger Coal	1884.30	[28]
Kiln outlet	687.9	[25,27]
Water stream to RC	4187	[25]
Waste gas	1140	[27]

Table 2
Operating cost of water and coal.

Material	Cost	
	Value	Unit
Iron ore	2600	Rs/tonne
Coal (Mixing of feed coal and Slinger coal)	2500	Rs/tonne
Water	60	Rs/kilolitre
Electricity cost	3.5	Rs/kWh
Standard Cost of Boiler Turbine System	46800	Rs/kW
Standard Maintenance cost of Boiler Turbine System	0.2	Rs/kWh
Standard Maintenance cost of Ceramic Filter	171.93	Rs/(m^3/s)
Unit cost of power generation around the turbine	6.5	Rs/kW

gas, which exits the ABC at 994 °C, for power generation. On the other hand, in Option-2 heat of waste gas is used to preheat kiln feed, air and slinger coal.

3.1. Option-1

The waste gas having heat capacity, CP, of 33.08 kW/°C consists of 24610.47 kW heat while moving from 994 °C to 250 °C as shown in Appendix A. This heat is recovered through power generation in Option-1. Waste gas at 994 °C enters to preheater, boiler and super heater and exits it at 250 °C.

3.2. Option-2

For Option-2, the solid streams such as iron ore, coal and slinger coal as well as gas streams like waste gas and air are used for heat integration. Preheating of feed materials, kiln air and slinger coal is carried out with the use of heat of waste gas. The stream data of Option-2 is given in Table 3. In the existing system waste gas cools down from 994 °C to 250 °C while moving through EC as shown in Fig. 1. This heat is used for the preheating of feed materials, slinger coal and kiln air. Therefore, the supply and target temperatures of waste gas are selected as 994 °C and 250 °C, respectively, as shown in Table 3. The target temperature as 200 °C is fixed for solid streams, i.e. feed material and slinger coal, due to process requirement as beyond this temperature the problem of conveying of solid may occur. Further, kiln air, which is the part of total air that enters to rotary kiln and is called as secondary air, is preheated up to 600 °C. In fact, in the kiln air, primary air enters to kiln with slinger coal is not included as coal may ignite beyond 200 °C before entering to kiln. Therefore, primary and secondary air are pre-heated up to 200 °C and 600 °C, respectively. The waste gas is involved with ABC unit, whereas kiln feed, kiln air and slinger coal are involved in rotary kiln.

4. Model for coal consumption

In the sponge iron process coal is the only source of energy, which produces through its combustion. The coal consumption is determined by energy demand of the process, which depends on

Table 3
Stream data of Option-2 for the modified system.

Steam	Type	T_s	T_t	Flow (kg/s)	C_p(J/kg/K)	CP (kW/°C)	Unit
Waste gas	h1	994	250	20.53	1140	23.402	ABC
Kiln air	c1	30	600	11.76	1032.6	12.148	Rotary kiln
Kiln feed	c2	30	200	10.90	1092.7	11.905	Rotary kiln
S coal	c3	30	200	1.38	1380	1.903	Rotary kiln

preheating of feed materials and air, heat involved in reduction reactions, heat lost through the kiln wall, preheating required by coal itself and latent heat required for evaporation of moisture of feed material. Detailed expression of coal consumption in derived in Appendix A.3.

5. Solution technique

For the solution of energy integration options, formulated in the present study, a method is developed and described as:

Step 1: The data is collected from existing plant as: (i) the process flow diagram (PFD) with operating parameters such as flow rate, temperatures, etc. as shown in Fig. 1, (ii) Physical properties of streams as given in Table 1.
Step 2: The overall mass and energy balance are carried out based on the existing process data.
Step 3: The air requirement is found based on the oxygen demand of the process, which is computed using different reactions involved in reduction as well as combustion process. The ratio of air to coal ({Air}/{Coal}) is calculated on the basis of data reported in the existing process, shown in Fig. 1.
Step 4: Energy conservation options are identified and corresponding stream data are extracted from exiting process.
Step 5: Process integration (PI) principles are applied on stream data to draw composite curve and grand composite curve. Power generation computation is carried out if required.
Step 6: The amount of coal for different options is computed using Eq. (10).
Step 7: Based on the ratio found in step 3, revised amount of air is calculated.
Step 8: Using new values of coal and air predicted in step 6 and 7, stream data is modified. For these data 2nd iteration is carried out while repeating Steps 5 to 6 till values of coal in two consecutive iterations are found equal.
Step 9: Economic analysis of energy conservation options are carried out. Under this operating cost of different commodities, capital cost of additional equipment, TAC, Profit and payback periods are computed.
Step 10: The feasibility analysis for these options are discussed.
Step 11: Based on economic analysis best option is selected.
Step 12: The modified PFD is prepared for best option using revised values of operating parameters.

6. Results and discussion

In the present work two energy integration options are proposed, which consider heat of waste gas that exits the after burning chamber (ABC) for energy integration. This waste gas is available from 994 °C to 250 °C. The results of these options are discussed hereunder:

6.1. Option-1

In the PFD of sponge iron process, shown in Fig. 1, waste gas carries 24610.47 kW of heat with heat capacity, CP, of 33.08 kW/°C when it moves from ABC to ESP. To recover this heat grand composite curve (GCC) is drawn as shown in Fig. 2 [18] considering ΔT_{min} as 50 °C. In fact, in energy integration ΔT_{min} is key parameter, which is to be optimized based on minimum total annual cost. For sponge iron process optimum ΔT_{min} was selected while varying it from 10 °C to 300 °C in the work of Prasad [19]. From this range optimum ΔT_{min} was found as 50 °C. As similar data is used for the present energy integration study, fixed value of ΔT_{min} is considered as 50 °C, which is already an optimum value for sponge iron process. Further, it is noted that as waste gas stream has lower heat transfer coefficient, it requires significantly larger area for heat transfer. However, the effect of lower heat transfer area can be somewhat compensated by increasing temperature difference (ΔT) between waste gas and cold stream. Thus, large ΔT is preferred for operation when gas stream is involved in heat transfer [18].

The GCC is temperature-enthalpy profile where temperature is shown at shifted scale. In Fig. 2 waste gas is drawn from 969 °C to 225 °C at shifted scale, which is 25 °C lesser than the actual temperatures i.e. 994 °C–250 °C. The steam profile is also plotted in GCC which includes preheating, boiling and superheating. For power generation it is assumed that saturated steam is produced at 280 °C whereas, superheated steam is generated at 350 °C and 70 bar. To draw steam profile initially water is preheated from 80 °C to 280 °C consuming 7876.2 kW of heat of waste gas as shown in Fig. 2. As the water stream takes heat from waste gas, it represents as cold stream. Thus, at shifted temperature, the water profile is heated up from 105 °C to 305 °C. Further, at 305 °C the profile is drawn as straight line as it is saturated steam, where preheated water is converted to saturated steam at 280 °C and 64.16 bar. Finally, superheated steam is produced at 350 °C and 70 bar, which is shown at 375 °C at shifted temperature scale in Fig. 2. The CP of preheated water and superheated steam are found as 39.38 kW/°C and 31.66 kW/°C, respectively. While boiling and superheating 14518.1 kW and 2216.14 kW heat, respectively, of waste gas are consumed. Consequently, waste gas drops its temperature to

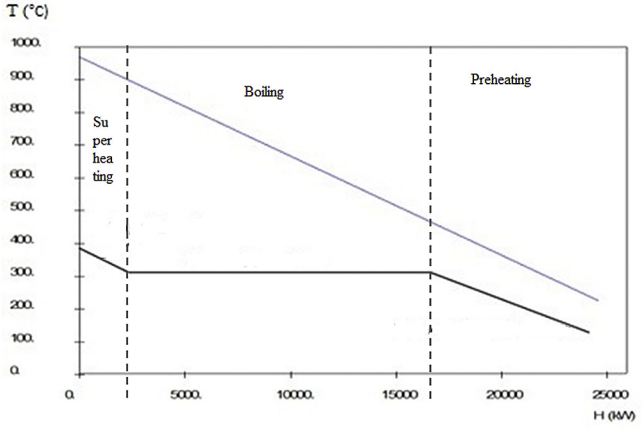


Fig. 2. Grand Composite curve for Option-1.

250 °C. In this process total 9.406 kg/s steam is produced. Considering 85% isentropic efficiency and 80 °C condensing temperature of superheated steam 6.88 MW of power is produced as shown in Table 4. The wetness fraction during the expansion is found as 0.155, which is almost equal to the limiting value i.e. 0.15 [20].

Further, effect of variation of three parameters such as saturation temperature, superheated temperature and superheated pressure on power generation is studied. For each variation total power generated, Carnot efficiency as well as wetness fraction are computed and shown in Table 4. Initially, saturated temperature is varied from 200 °C to 280 °C considering fixed values of superheated temperature and superheated pressure as 350 °C and 70 bar, respectively. The maximum power generated is found as 6.88 MW at 280 °C. The value of wetness fraction is found as 0.155, almost equal to the maximum limit i.e. 0.15. However, considering this factor, 280 °C is considered as optimum saturated temperature. Further, Carnot efficiency is found equal to 43%. Then, superheated temperature varies from 300 °C to 700 °C keeping constant values of saturated temperature and superheated pressure as 280 °C and 70 bar, respectively. The selected maximum power generation is 7.67 MW corresponding to 600 °C, where wetness fraction and Carnot efficiency are predicted as 0.003 and 59%, respectively. Finally, superheated pressure is varied from 80 bar to 400 bar while fixing values of saturated temperature and superheated temperature at optimum conditions.

For this case maximum power of 9.13 MW is generated with wetness fraction of 0.168 and Carnot efficiency as 57%, which are corresponding to superheated pressure of 400 bar. As wetness factor achieved at 400 bar is beyond the tolerable value of 0.15 the optimum superheated pressure is accounted as 350 bar where power generation is 8.97 MW and wetness factor is 0.147. The final cycle of power generation with all operating parameters is shown in Fig. 3.

For Option-1, coal consumption is same as used in the existing PFD shown in Fig. 1, which is 21.72 t/h. This is due to the fact that using heat of waste gas in power generation the operation in rotary kiln is not affected and thus, the coal consumption. On the other hand, water consumption in the modified system is changed than that is supplied in the existing PFD and revised amount of water consumption is found as 1242.7 t/h. In the existing PFD, Fig. 1, water is consumed in two units, rotary cooler and EC, whereas in the modified system it is required in rotary cooler and power generation cycle. The water consumption in rotary cooler is 750 t/h in both systems. However, in the modified system 29.56 t/h water is consumed in steam cycle whereas, 463.32 t/h (equal to 128.7 kg/s) is used in condenser as shown in Fig. 3, which shows total water requirement as 1242.7 t/h. Thus, 56.3% more water is used in the modified system in comparison to the existing PFD, where water consumption is 795 t/h.

6.1.1. Operational and economic details for Option-1

Waste gas that exits the rotary kiln consists of 11.5 t/h of dust. A part of it is settled in DSC and ABC and 6.1 t/h dust is present in waste gas, when it comes out from ABC. A ceramic filter is proposed for this purpose, which is installed between ABC and power generation system. Once this filter is used the problem of scaling or dust deposition in the tubes of boiler and other exchangers can be somewhat compensated, which enhances operability of power generation system. However, 8.21 kg/s of water with CP 34.37 kW/°C is preheated from 80 °C to 280 °C using 6873.6 kW heat of waste gas. During this exchange waste gas drops its temperature from 457.8 °C to 250 °C. Heated water then enters to boiler where saturated vapor is formed at 280 °C and 64.202 bar pressure. In this process waste gas releases 12669.3 kW of heat and so, its temperature is decreased from 840.8 °C to 457.8 °C. Further, saturated vapor enters to super-heater where it is heated up to 600 °C and 350 bar pressure while consuming 5066.95 kW of heat of waste gas. Consequently, temperature of waste gas drops from 994 °C to 840.8 °C. Therefore, while consuming 24610.47 kW of heat of waste gas water rises its temperature from 80 °C to 600 °C. Further, supplying same amount of heat of waste gas to power generation system, it exits the system at 250 °C and then enters to ESP as carried out in the existing PFD as shown in Fig. 1. Therefore, installing ceramic filter as well as power generation system does not affect the operation of sponge iron production process. Moreover, the suggested scheme for recovery of heat of waste gas

Table 4
Variation of parameters for Power Generation.

S. No.	Saturation temperature (°C)	Superheated temperature (°C)	Superheated pressure (bar)	Power (MW)	Stream flow rate, kg/s	Wetness fraction
1	280	350	70	6.88	9.41	0.155
	270	350	70	6.86	9.37	0.155
	260	350	70	6.84	9.34	0.155
	240	350	70	6.80	9.29	0.155
	230	350	70	6.79	9.27	0.155
	220	350	70	6.77	9.25	0.155
	200	350	70	6.75	9.23	0.155
2	280	300	70	6.76	10.09	0.206
	280	400	70	7.02	8.91	0.118
	280	450	70	7.17	8.52	0.086
	280	500	70	7.33	8.17	0.056
	280	550	70	7.49	7.85	0.028
	280	600	70	7.67	7.57	0.003
	280	700	70	8.02	7.05	0.046
3	280	600	80	7.79	7.58	0.012
	280	600	90	7.89	7.61	0.021
	280	600	100	7.99	7.63	0.029
	280	600	125	8.18	7.68	0.046
	280	600	150	8.34	7.73	0.061
	280	600	175	8.46	7.78	0.074
	280	600	200	8.57	7.84	0.086
	280	600	250	8.73	7.95	0.108
	280	600	300	8.85	8.08	0.127
	280	600	350	8.97	8.21	0.147
	280	600	400	9.13	8.33	0.168

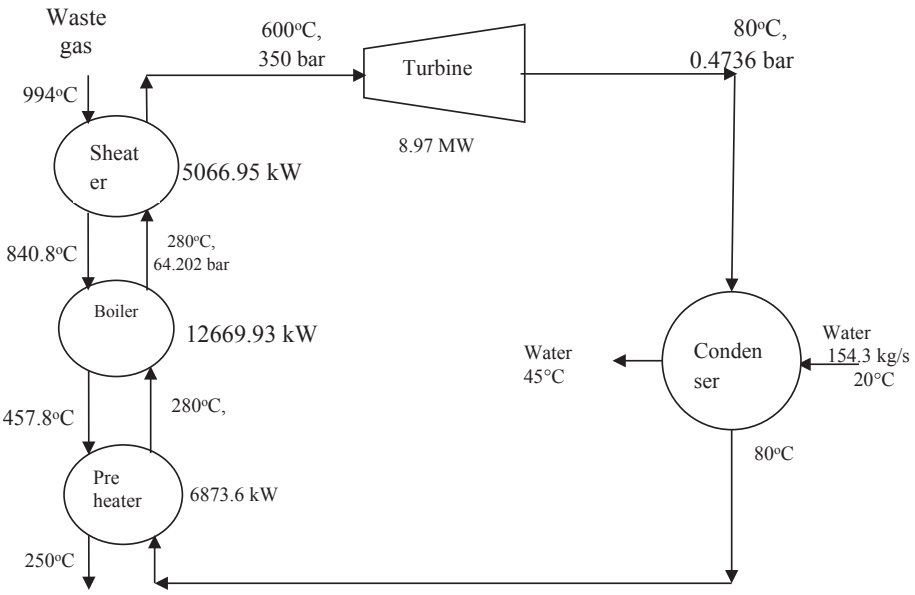


Fig. 3. Power generation cycle.

through Option-1 is applied once the waste gas exits the rotary kiln. Thus, using Option-1 process inside the rotary kiln will not be disturbed. The economic analysis of Option-1 accounts the operating as well as capital costs as shown in Table 5.

6.1.1.1. Operating cost. The operating cost of Option-1 depends on coal and water requirements. It is changed only due to water consumption, as no change in coal consumption is found. The costs of coal and water are taken from Table 2. It is computed considering 8000 h per year as working hours of plant. Based on these values total operating cost for Option-1, is found as Rs 11268.8 lakh/year in which Rs 4344 lakh/year and Rs 5964.9 lakh/year are due to coal and water consumption. It does not include costs of iron ore and dolomite as these are fixed quantities for all options and thus can be excluded from comparative study. The operating cost of Rs 816.3 lakh per year is predicted for maintenance of the filter based on the volume of dust gas handled by the ceramic filter per unit time. Due to steam generations there is regular wear and tear in parts of boiler turbine system. This causes an expenditure of Rs 143.52 lakh per year in maintenance.

The ceramic filter used in this option costs Rs 4157.3 lakh [21], which is used to handle 87048.92 m^3/h of waste gas. Based on this total capital investment in boiler and turbine system is found as Rs 4197.96 lakh. Considering straight line depreciation with 10 years as equipment life the total annual cost (TAC) is computed as Rs 12104.3 lakh/year. Also, total annual profit is computed considering revenue generated while exporting the power, expense of additional water requirement and maintenance costs of filter and power generation system and it is found as Rs 2029.9 lakh/year. Thus, the payback period of 4.17 years is predicted for Option-1 as shown in Table 5.

6.1.2. Feasibility analysis of Option-1

Waste gas consists of significant amount of dust, which may be deposited on the surface of tubes in boiler, preheater and superheater. Due to continuous deposition of dust the operation inside these become infeasible. This problem can be handled in two ways: firstly, the equipment should be cleaned periodically to remove the deposited dust and secondly, dust is removed from waste gas before it enters to power generation system. First option requires continuous cleaning and maintenance of equipment whereas, in second case comparatively clean gas will enter the system and so, less maintenance of the equipment is required. Therefore, in the present work second case is selected and for this purpose high temperature ceramic filter is proposed. It traps the dust particles and releases comparatively clean gas.

Further, in the modified PFD movement of waste gas is changed in comparison to the existing PFD, shown in Fig. 1, as power generation system is installed in between the path of waste gas. To make this movement feasible an induced draft fan is used before chimney. In the existing system this fan is already installed. However, in the modified PFD induced draft fan with larger capacity is

Table 5
Economic analysis of Option-1.

Operating cost (lakh/year)			Capital cost (lakh)		Power produced (MW)	Power Exported (lakh Rs)	TAC (lakh/year)	Profit (lakh/year)	Payback period (Years)
Commodity	Amount, t/h	Cost	Item	Cost					
Coal	21.72	4344	Boiler and turbine system	4197.96	8.97	5107.52	12548.8	1591.8	5.32
Water	1335.3	6409.4	Ceramic filter	4157.3					
Maintenance cost of ceramic filter		816.3							
Maintenance cost of turbine system		143.52							

Table 6
Results of iterations for computing coal consumption in Option-2.

Iteration no.	Coal consumption, t/h	Air requirement, t/h	Flow rate of waste gas, t/h
1	21.720	84.063	143.4
2	13.587	52.588	112.8
3	13.587	52.588	112.8

required as waste gas moves through more complicated path as compared to the existing system.

6.2. Option-2

At first, waste gas is used to preheat feed material from 30 °C to 200 °C having CP of 13.364 kW/°C. Then the remaining heat available with waste gas is used to preheat kiln air from 30 °C to 600 °C, which has CP value of 18.917 kW/°C. Further, slinger coal is preheated from 30 °C to 200 °C. During these exchanges waste gas drops its temperature to 584.12 °C. As a result of preheating of feed material, kiln air and slinger coal waste gas acquires temperatures of 925.32 °C, 599.35 °C and 584.12 °C, respectively. The temperature 200 °C is fixed for solid streams, due to process requirement as beyond this temperature the problem of conveying of solid may occur. Considering these temperatures of different streams, stream data is prepared for Option-2 as presented in Table 3. Considering ΔT_{min} as 50 °C, Pinch analysis is applied to the stream data. The hot and cold utilities as 0 kW and 11052.3 kW, respectively, are found. It is a threshold problem as the hot utility requirement is zero.

Coal consumption is reduced significantly due to preheating as coal is utilized to heat air, iron ore and coal from 600 °C, 200 °C and 200 °C, respectively, to reaction temperature as shown in existing system, Fig. 1. In fact, in the existing sponge iron process 21.720 t/h and 84.063 t/h of coal and total air, respectively, is supplied, which is in a ratio of 3.87. It is to be maintained in the process. The coal consumption for Option-2 is found as 13.587 t/h using Eq. (7). It is 37.4% less in comparison to the existing system shown in Fig. 1. Corresponding to new value of coal the revised flow rate of air is found as 52.588 t/h. The new values of CPs of waste gas, kiln air, feed material and slinger coal are 23.402 kW/°C, 12.148 kW/°C, 11.905 kW/°C and 1.903 kW/°C, respectively. These values are used to modify the stream data shown in Table 3. Pinch analysis is applied to modified stream data to find revised values of hot and cold utility, which are found as 0 kW and 8138.728 kW, respectively. The results of all iterations are summarized in Table 6. During this integration waste gas drops its temperature to 597.8 °C and further reduction in the temperature to 250 °C cold utility of 8138.7 kW is required. Water at 30 °C is used as cold utility, which is supplied in EC to cool waste gas from 597.8 °C to 250 °C. For this purpose, 14.882 t/h water is used. However, 750 t/h water is also used in rotary cooler to cool kiln outlet stream as it is carried out in the existing system, which shows total water consumption as 764.882 t/h, which is 3.8% less as compared to the existing system. Further, it is noted from Table 6 that reduced amounts of coal and air cause less waste gas to discharge from the process. The waste gas is reduced from 143.4 t/h to 112.8 t/h, which is an additional benefit of energy integration through Option-2.

The heat exchanger network (HEN), which utilizes 13.587 t/h of coal is shown in Fig. 4. It shows that to achieve possible recovery of heat three new heat exchangers are required. Amongst these two are gas-solid and one is gas-gas type exchangers.

6.2.1. Operational and economic details for Option-2

In Option-2, waste gas consists of 6.1 t/h dust when it comes out from ABC. To remove these dust particles a high temperature ceramic filter is installed, through which operability and maintenance of preheater become easier. Further, an induced draft fan is placed before the chimney which helped in the movement of waste gas from rotary kiln to chimney. However, for preheating process streams waste gas should traverse from ABC to rotary kiln, which is in backward direction. For this movement of gas forced draft fans are used in gas carrying ducts.

Waste gas is cooled from 994 °C to 907.5 °C while supplying its heat to feed material as shown in Fig. 4, through insulated duct of 25 m length from ABC to kiln inlet. To preheat the feed material direct contact gas-solid heat exchanger of 2023.9 kW is employed, which is of rotary cylinder type. Further, waste gas at 907.5 °C is used to preheat kiln air. For this purpose, waste gas traverses 80 m distance from kiln inlet to kiln outlet using a gas carrying insulated duct to reach to the preheater. A shell and tube heat exchanger is selected, which is of gas-gas type. As waste gas has more fouling tendency, it is allocated to tube side and thus, air is entered to shell side. Considering overall heat transfer coefficient as 50 W/m^2°C [22] for exchanger heat transfer area is predicted as 322 m^2.

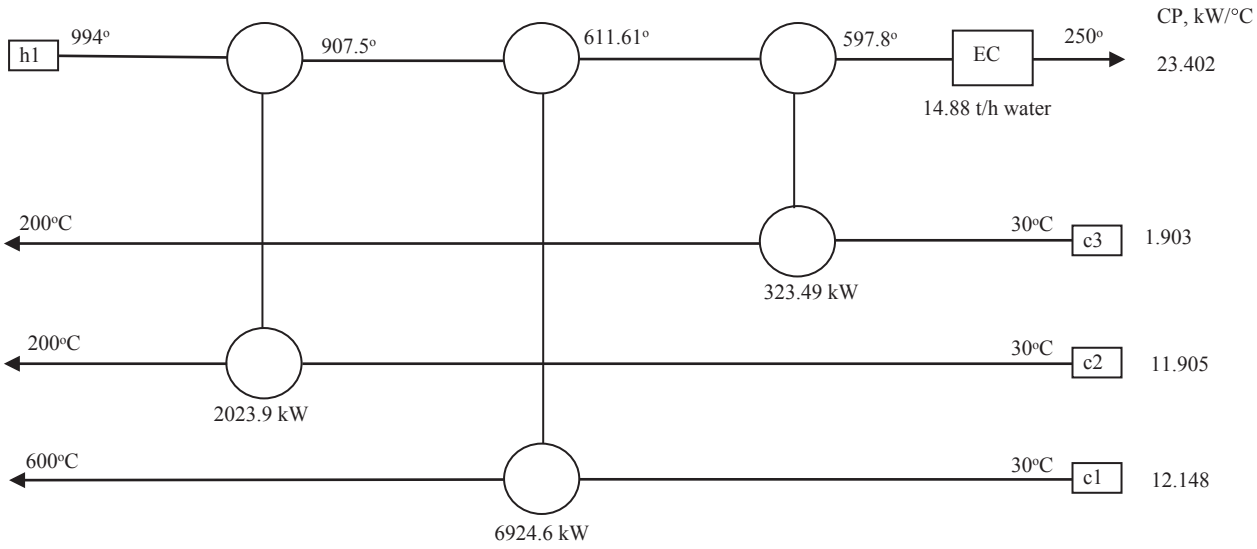


Fig. 4. HEN diagram of Option-2.

Further, gas at 611.61 °C releases its heat to preheat slinger coal. The heated slinger coal goes to the kiln from discharge end. Therefore, waste gas exits the air preheater can directly enter to it at 611.61 °C. It is a gas-solid heat exchanger of 323.49 kW heat duty, where slinger coal is heated from 30 °C to 200 °C.

The waste gas at 597.8 °C enters to non-insulated duct of 105 m length to move form kiln outlet to EC inlet. In the EC temperature of waste gas is dropped to 250 °C using cooling water of 14.882 t/h. As a result, water consumption will be decreased in EC from 45 t/h to 14.88 t/h. Therefore, in Option-2 total three ducts are required. The casing of these ducts is made of mild steel. However, for insulated ducts outside insulation of ceramic fiber of thickness of 100 mm is used. The temperature profiles for 25 m, 80 m and 105 m long ducts are shown in Fig. 5. The exit temperatures of gas from 25 m, 80 m and 105 m long ducts are 992.72 °C, 902.5 °C and 432.1 °C, respectively, which is shown in Fig. 6. At 432.1 °C waste gas enters to EC where it achieves 250 °C using 7.8 t/h water. Therefore, using duct maximum heat of waste gas is lost through duct and remaining heat is released using water. Thus, Option-2 utilizes 757.8 t/h water instead of 764.882 t/h.

Further, the pressure drops for ducts are obtained using following equation [23]:

$$\Delta P = \left(0.109136q^{1.9}\right)\Big/ de^{5.02} \tag{8}$$

The pressure drops are found as 0.034 atm, 0.12 atm and 0.16 atm for 25 m, 80 m and 105 m long ducts, respectively. One FD fan is used in first duct of 25 m length to sustain 0.034 atm pressure. Similarly, duct of 80 m and 125 m length require one FD fan each for maintaining necessary pressures in the duct. Thus, modified PFD for Option-2 requires three FD fans.

The economic analysis of Option-2, in terms of operating cost and capital cost, is shown in Table 7. The payback period of 3.22 years is predicted for Option-2.

6.2.2. Feasibility analysis of Option-2

The feasible solution for removing dust from waste gas is to install a high temperature ceramic filter after the ABC as it is used in

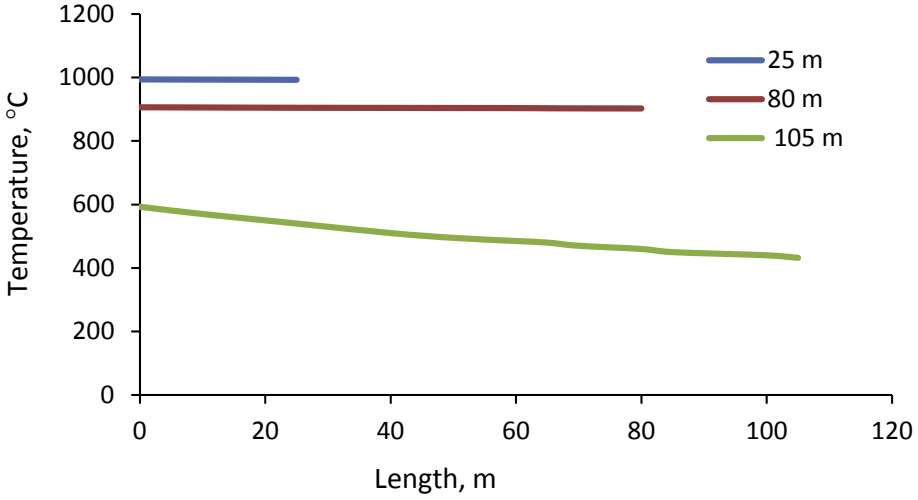


Fig. 5. Duct temperature profile along the length of the kiln.

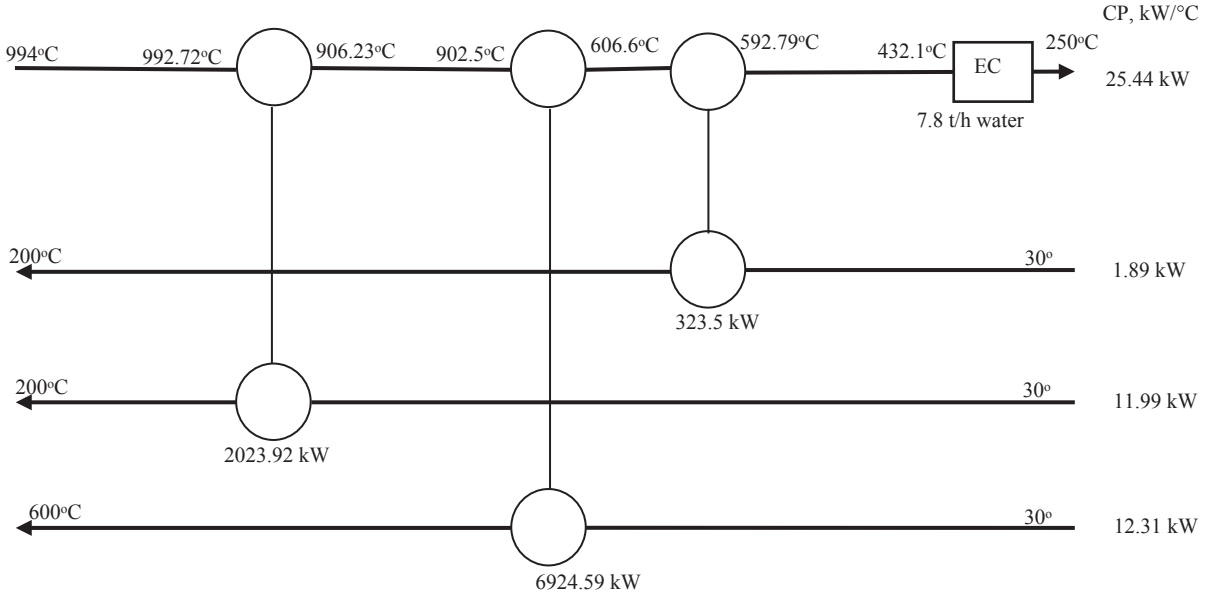


Fig. 6. Modified HEN diagram of Option-2.

Table 7
Economic analysis of Option-2.

Operating cost (lakh/year)			Capital cost (lakh)		TAC (lakh/year)	Profit (lakh/year)	Payback period, year
Commodity	Amount	Cost	Item	Cost			
coal	13.58	2678.1	G-G HX	50.8	7272.4	1126.8	3.22
water	757.8	3584.7	G-S HX	142.14			
Maintenance cost of ceramic filter		577.5	G-S HX	22.72			
Electricity cost		74.5	Ceramic filter	2941.1			
			DUCTS	8.97			
			INSULATION	42.79			
			FD fans	367.26			

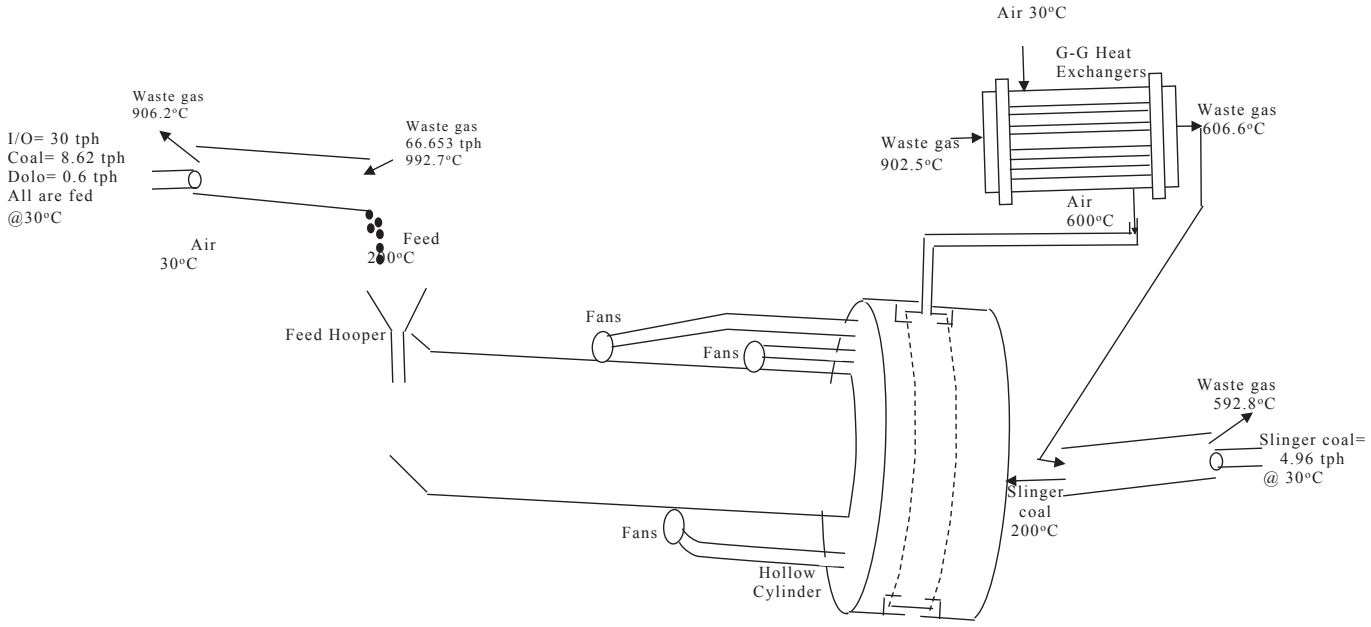


Fig. 7. Feasibility Diagram for option-2.

Option-1. The filter is shown in Fig. 7, which is used in modified PFD. For the purpose of preheating waste gas moves through different ducts. This movement is against the usual flow of waste gas, which can be made feasible through FD fans placed at different ducts.

Firstly, the waste gas preheats the feed material up to 200 °C in a rotary direct contact gas-solid heat exchanger placed near the rotary kiln as shown in Fig. 7. In this exchanger feed material enters at 30 °C direct through the conveying belt. Waste gas also enters to the exchanger from feed side. Due to heat exchange, heated feed materials, directly goes to hopper and then to rotary kiln. Waste gas exits the gas-solid exchanger at 992.72 °C.

Further, it traverses through 25 m length duct to reach to discharge end of kiln where gas goes into shell and tube heat exchanger at 906.23 °C. As it moves through tubes of exchanger and cools to 902.5 °C. Consequently, air is heated from 30 °C to 600 °C, which is to be entered at 600 °C in rotating kiln. The hot air exits the shell and tube exchanger and directly goes to hollow cylinder chamber is placed at the discharge end of kiln. As hot air will enter through nine blowers, these are attached with nine ducts. Other sides of these ducts are connected to the hollow cylinder through which hot air continuously enters to the kiln at 600 °C. The flow rate of air is maintained by metal dampers present in the blowers, which can sustain with high temperature of air i.e. 600 °C. To avoid air to be entered at atmospheric temperature in the kiln through nine blowers one should ensure that the hollow cylinder must be filled with hot air always. To ensure it, a specific assembly is proposed as illustrated through Fig. 7, where a circular plate of metal is placed inside the hollow cylinder. In this plate outlet of hot air pipe, which exits the shell and tube exchanger, is attached. The circular plate and hot air pipe remain static. Further, a U-shape assembly is proposed which rotates with hollow cylinder and in between the U-shape assembly circular plate is present. This plate works as sealing and avoids mixing of hot air with atmospheric. The assembly of hollow cylinder as well as nine ducts rotate continuously with rotary kiln at same rpm and allow hot air to be entered in the kiln. However, the shell and tube heat exchanger and duct between it and circular plate remain stationary and thus, it does not obstruct the operation of kiln.

Once the waste gas exchanges heat with air, it is used to preheat slinger coal through direct contact gas-solid heat exchanger. The design and operation of this exchanger is similar to that employed for preheating of iron ore. It is placed near the discharge end of kiln as shown in Fig. 7. In this exchanger slinger coal enters at 30 °C through conveyer and heats up to 200 °C. Waste gas at 592.79 °C goes into this exchanger and exits it at 432.1 °C. The heated slinger coal goes through the kiln from discharge end pneumatically. Further, waste gas moves through 105 m length duct to reach to EC at 432.1 °C where it is cooled to 250 °C using 7.8 t/h of water.

6.2.3. Comparison of results of Option-1 and Option-2

In the present work, two options have been examined to find

out the best scenario. The values of economical parameters such as coal consumption, water consumption, operating cost, payback period, capital cost and waste gas generation for Option-1 and Option-2 are compared in Table 8.

The maximum coal consumption can be saved by heat integration is 37.4%, which is found for Option-2. The water consumption in Option-2 is 37.2 t/h less than the existing case where it is 795 t/h. On the other hand, it is 447.7 t/h more for Option-1. Thus, minimum operating cost is found for Option-2, which is 39.25% less in comparison to Option-1 as shown in Table 8. It shows that the variation in capital cost is in the following order: Option-1 > Option-2. The payback periods for these options are found in the order: Option-1 > Option-2. The difference between maximum and minimum payback periods, is 0.95 years or 11.4 months. It is clear from Table 8 that waste gas produced in Option-2 is 47.6% less in comparison to existing case. The performance of each option can be estimated by determining the profit as well as investments, which are determined in terms of fixed cost and operating cost of the items. Thus, Option-1 gives maximum profit and also involves maximum investment. The total annual cost (TAC) is minimum for Option-2. Based on these values, Option-2 is considered as the best energy conservation option. The modified PFD of Option-2 with revised values of coal and water is shown in Fig. 8. The total theoretical energy used for the reduction of 30 tonnes ore is found as 1.16×10^7 kJ/kg using heat of reaction of reduction reactions to convert Fe_2O_3 to Fe [24,25]. The actual energy requirement is computed while multiplying coal consumption and net heating value. In the case of Option-1 or exiting system, the actual energy consumed is found as 4.94×10^8 kJ/kg which is 42.43 times more than the theoretical value. Similarly, in the case of Option-2, the actual energy consumed is found as 3.09×10^8 kJ/kg which is 26.64 times more than the theoretical value. Thus, Option-2 is more close to the theoretical value.

6.2.3.1. Uncertainty analysis. In the present work, a simulation work is carrying out using the plant data of an existing sponge iron plant and it shows that the results obtained through simulation are found to be correct. Thus, it is clear that the uncertainty analysis may not be useful in this simulation based analytical study as it is more applicable to experimental work where a number of different measurements of different quantities are carried out to determine certain parameter [19].

To illustrate above facts a few cases are discussed as: In the present work coal consumption, predicted using Eq. (10), depends on mass, specific heat capacity and temperature of input streams and error observed in predicting these parameters may cause uncertainty in estimating coal consumption. As, in the present work, values of these parameters are directly taken from literature as well as plant, error is not found in these values and thus, uncertainty in predicting coal consumption may not be found at present state of computation. In the similar lines, uncertainty in cost of a heat exchanger can be discussed considering G-G heat exchanger of the best case, Option-2. The cost of this exchanger, predicted using correlation of Shenoy [26], depends on area of G-G heat exchanger. The area is computed based on heat load of exchanger, log mean temperature difference and U. Once the heat exchanger network is designed heat load as well as log mean temperature difference of a exchanger is fixed. Moreover, the value of U is taken from literature [27]. Considering these values area is computed as 325.93 m^2 where uncertainty cannot be determined.

Apart from it an effort is made to determine uncertainty in the total annual cost (TAC) of Option-2. The TAC of a scenario is predicted as:

$$TAC = CC + OC \tag{9}$$

The maximum value of uncertainty in TAC is given as:

$$U_{TAC} = \left[\left(\frac{\partial TAC}{\partial CC}\right)^2 (U_{CC})^2 + \left(\frac{\partial TAC}{\partial OC}\right)^2 (U_{OC})^2\right]^{1/2} \tag{10}$$

or

$$U_{TAC} = \left[(U_{CC})^2 + (U_{OC})^2\right]^{1/2} \tag{11}$$

To illustrate the uncertainty analysis an example of G-G heat exchanger of 325.93 m^2 area, proposed for Option-2, is considered. In the present work cost correlation of Shenoy [26] is used to compute capital cost of this heat exchanger. As this correlation is considerably old (i.e. 1995) it cannot give recent cost of exchanger which may cause uncertainty in TAC. For G-G exchanger capital cost is computed as 50.8 lakh using correlation of Shenoy [26] as shown in Table 7. Consequently, annualized capital cost is 5.08 lakh considering life of heat exchanger as 10 years. However, for the same area of heat exchanger annualized cost predicted using new method shown in Ref. [28] is found as 5.93 lakh. So, uncertainty in annualized capital cost is found as +0.85 lakh. The uncertainty in annual operating cost is negligible as it depends on cost of coal, water and electricity, which are taken from the plant. Thus, Eq. (11)

Table 8
Comparison of different heat recovery options.

Option	Coal cons., t/h	Water cons, t/h	Operating cost, lakh/year	Electricity cost, lakh/year	Capital Investment		TAC, lakh/year	Waste gas generated, t/h	Profit, lakh/year	Payback period, Years	Power Generation, MW
Existing System	21.72	807	8172.9	–				157.04			–
1	21.72	1335.3	11713.3	–	Item	Cost, lakh	12548.8	157.04	1591.8	5.32	8.97
					Boiler & turbine system	4197.96					
					Ceramic filter	4157.3					
2	13.587	757.8	6914.9	74.5	G-G HX	50.8	7272.4	112.8	1126.8	3.22	–
					G-S HX	142.14					
					G-S HX	22.72					
					Ceramic filter	2941.1					
					Ducts	8.97					
					Insulation	42.79					
					FD fan	367.26					

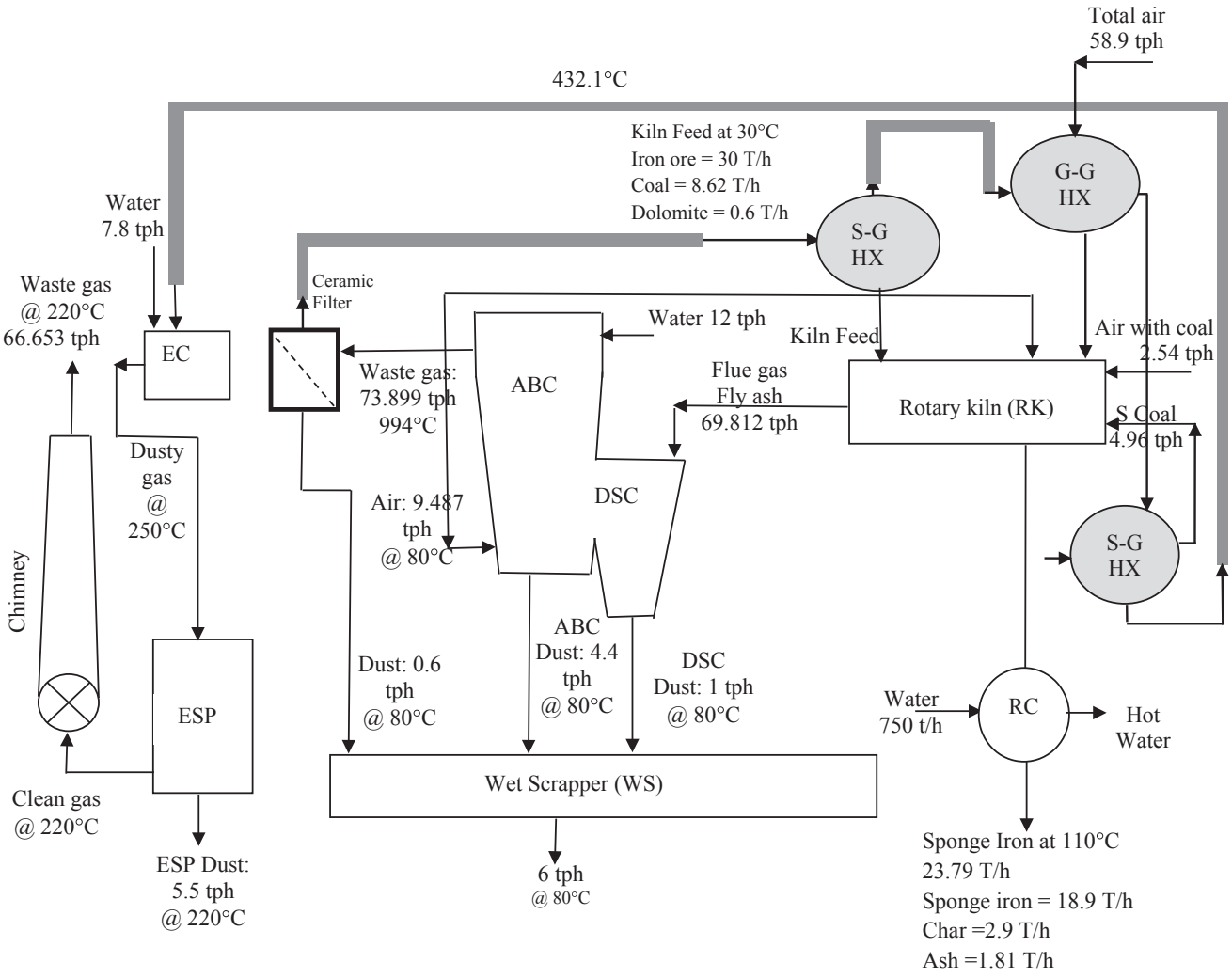


Fig. 8. Modified PFD of the process.

is reduced to

$$U_{TAC} = U_{CC} \tag{12}$$

The uncertainty in TAC is found as +0.85 lakh due to uncertainty in annualized capital cost. Thus, TAC of Option-2 is increased by 0.015%. It increases the payback period by 15 days which is not significant.

6.2.4. Comparison of results of best option with that of published work

The results of best modification selected in the present work, Option-2, are compared with that of published work such as Prasad et al. [22], Prasad et al. [10], Kumar and Khanam [9] as shown in Table 9. For this purpose, total required capital, coal and water consumptions, payback period, profit, etc. are compared.

Prasad et al. [22] proposed eight cases for energy conservation. The best case consumes 93.7% and 12.3% less water and coal as compared to the existing system. Three S-G and one G-G heat exchangers are required in this case with an investment of Rs. 621.06 lakh. In Option-2 of the present work similar case study is chosen for heat integration. Option-2 is compared with the best case of Prasad et al. [22] and found that air is preheated up to 600 °C in Option-2 instead of 300 °C in Prasad et al. [22]. Due to higher temperature of preheated air in Option-2 it recovers more energy in comparison to the best case of Prasad et al. [22]. Therefore, reduction of coal in Option-2 and Prasad et al. [22] are found as 37.4% and 12.3%, respectively, as shown in Table 9. Though coal saving in Option-2 is higher than that of Prasad et al. [22], the payback period in Prasad et al. [22] is very less as compared to Option-2. This is due to the fact that in Prasad et al. [22] 93.7% saving of water was observed whereas it is only 4.7% in Option-2, which increases profit for Prasad et al. [22] many folds in comparison to Option-2 as shown in Table 9. However, Prasad et al. [22] did not discuss the feasibility of modification proposed while reducing 93.7% water. They proposed that waste gas was used as a coolant in rotary cooler instead of water. However, using waste gas in rotary cooler is not feasible as cooler is rotating device and waste gas is coming from a stationary source such as ESP. Thus, practical implementation of best case of Prasad et al. [22] is not clear. On the other hand, in Option-2 water is used to cool kiln outlet in cooler, which is more feasible solution. Further, ceramic filter is also used as additional equipment along with different heat exchangers as well as FD fans. The cost of ceramic filter is significantly higher in comparison to other equipment cost as shown in Table 8. Thus, due to higher capital investment and lesser profit than that of Prasad et al. [22] payback period is higher for Option-2.

Net profit of a heat recovery option is computed by multiplying total profit with plant life minus payback period. The payback

Table 9
Comparison of results of Option-2 with the published work.

S. No.	Strategy Name/ Author name	Integration done in best case	Capital required, lakh Rs	Coal saved %	Water saved %	Payback Period, years	Profit, lakh Rs/year	Net Profit in plant life, lakh Rs
1	Prasad et al. [21]	Preheating of air up to 300 C with ESP exit	621.06	12.3	93.7	0.15	3092.6	30456.79
2	Prasad et al. [10]	Preheating of feed material and air up to 110 C and 385.7 C	588.2	30.5	72.6	0.56	1180.2	11141.09
3	Kumar and Khanam [9]	Air is preheated to 80 C by kiln outlet using water bath	122.98	4.8	96.3	0.93	132.9	1206.07
4	Option-2 with preheating	Air preheated to 600 C by kiln outlet. Feed and slinger coal preheated to 200 C by ESP exit.	3575.7	37.4	4.7	3.22	1126.82	7639.84

periods for the options are predicted based on capital investment of the modification and total profit. The plant life considered in present work and Prasad et al. [22] is 10 years. Due to very less payback period for Prasad et al. [22] the net profit is significantly higher than that of Option-2.

Prasad et al. [10] proposed two modifications for the utilization of heat of waste gas. These cases are compared with the existing system on the basis of coal and water consumption, capital cost, profit, and payback period and Case-2 is found as the best modification where preheating of air and feed material is carried out up to 385.7 °C and 110 °C, respectively. While comparing the results of Case-2 with that of Option-2 of the present study, it is found that for Option-2 preheating of air, feed material and slinger coal is done up to 600 °C, 200 °C and 200 °C, respectively. Consequently, saving of coal is found as 37.4% for Option-2, whereas it is only 30.5% in Case-2 as shown in Table 9. The payback period in Option-2 is significantly high as compared to Prasad et al. [10]. This is due to 6.1 times higher capital investment in Option-2 than that of Case-2 of Prasad et al. [10]. Further, Case-2 consumes 72.6% less water, whereas it is reduced by 4.7% in Option-2. As a results of it, profit increases for Case-2. Thus, more capital investment and less profit causes more payback period in Option. However, practical implication of modification in Case-2 is not discussed by the authors, which is the main drawback of the study.

Based on the plant life of 10 years, the net profit of the plant life is higher than that of Option-2.

Kumar and Khanam [9] also proposed two cases for the recovery and utilization of heat of waste gas. The best case of Kumar and Khanam [9] consumes 4.8% less coal, which is 37.4% in Option-2. It is due to preheating of air up to 600 °C, which is significantly higher than that of best case of Kumar and Khanam [9], where air is preheated only up to 80 °C as shown in Table 9. For Option-2 payback period is very high than that of Kumar and Khanam [9] due to similar facts as discussed for Prasad et al. [22]. Further, it is observed from Table 9 that profit is significantly higher for Option-2 than that of Kumar and Khanam [9] as Option-2 saves huge amount of coal in comparison to best case of Kumar and Khanam [9]. Due to more profit net profit in the plant life is higher for Option-2 as compared to Kumar and Khanam [9] as shown in Table 9.

Based on above discussion it is found that results of Option-2 are compared well with that of published work. In comparison to a few studies such as Prasad et al. [22] and Prasad et al. [10] Option-2 does not show higher profit than these. However, the modifications proposed in these studies are not supported by the feasibility analysis and thus, its practical implications are questionable. On the other hand, practicality of Option-2 is discussed in detail in the present study.

Further, the gap between theoretical and actual energy requirement is reduced by 0.9 times and 0.67 times in the case of Prasad et al. [22] and Prasad et al. [11], respectively. Thus, Option-2 saves more energy than that found through published work.

7. Conclusions

In the present work two heat integration options are proposed which utilizes heat of waste gas available in the sponge iron process. Option-1 consumes heat of waste gas in power generation whereas, that is consumed in preheating of kiln feed and air in Option-2. The salient conclusions of the study are:

1. Based on payback period preheating is more profitable option in comparison to power generation. However, when net profits of two options are compared, power generation option is more profitable.
2. Based on heat integration, Option-2 is selected as the best option among two. It saves 37.4% coal and 4.7% water. Further, waste gas is reduced by 21.3% in Option-2, which is an additional benefit.
3. The gap between theoretical and actual energy consumption is reduced significantly in Option-2 as compared to the published work. Further, the results of the present study are compared well with that of published work.

Conflicts of interest

None declared.

Appendix A

A.1. Empirical correlations for heat capacities of substances

The empirical correlations for heat capacities of different substances are [29].

Fe_2O_3: $C_p = 24.72 + 0.01604 \times T - 423400/T^2$

CO: $C_p = 6.6 + 0.0012 \times T$

CO_2: $C_p = 10.34 + 0.00274 \times T - 195500/T^2$

FeO: $C_p = 12.62 + 0.001492 \times T - 76200/T$

Fe: $C_p = 6.12 + 0.00336 \times T$

C: $C_p = 2.673 + 0.002617 \times T - 116900/T^2$

The calculation of Cp value for kiln feed is:

The specific heat of kiln feed depends on the composition of kiln feed which consists of iron ore, coal and dolomite. The specific heats of iron ore, feed coal and dolomite are 960, 1380 and 1028.38 J/kg°C [29–31] whereas, mass flow rates of iron ore, feed coal and dolomite are 30000 kg/h, 13990 kg/h and 600 kg/h, respectively.

Specific heat of kiln feed (J/kg°C) = [(specific heat of the component × mass flow rate of the component)/Total mass flow rate of all components]

Thus, = [{(30000 × 960)+(13990 × 1380)+(600 × 1028.38)}/{(30000 + 13990+600)/1000}]

= 1092.7 J/kg/°C

The specific heats of other streams are shown in Table 1.

A.2. Heat of reaction

Expression of heat of reaction is

Reaction: $Fe_2O_3 + CO = 2\ FeO + CO_2$

Fe_2O_3 Present in Iron Ore = {Iron Ore feed rate × % of Fe (Total) in Fe_2O_3 × Ratio of Fe_2O_3/Fe_2 }

= {30000 × 0.63 × ((2 × 56 + 3 × 16)/(2 × 56)) = 27000 kg

Fe_2O_3 lost in the Fly Ash = Production of Fly Ash × % of Fe_2O_3 in Fly Ash = 3159 × 0.424

= 1339.4 kg

Fe_2O_3 reacted = 27000−1339.4 = 25660.6 kg

Moles of Fe_2O_3 = Weight of Fe_2O_3/Molecular weight of Fe_2O_3 = 25660.6/160 = 160.3787

The enthalpy of FeO from 30 °C to 994 °C

Enthalpy of FeO = $mC_p\Delta T$

C_p = 12.62 -[(0.001492 × (1267 + 303))/2]-[76200/(1267 × 303)] = 11.2503 kCal/Kkmol

where, 1267 and 303 temperature corresponds to 994 °C and 30 °C respectively.

Thus, enthalpy of FeO = ((2 × 160.3787 × 11.2503 × (1267−303))/1000000 = 3.478 GCal

The enthalpies of the CO_2, Fe_2O_3 and CO are found as 1.852 GCal, 5.598 GCal and 1.166 GCal respectively.

The heat of formations for Fe_2O_3, CO, FeO and CO_2 are taken from Green and Perry [25] and found as −198.5, −26.4157, −64.62 and −94.0518 kCal/mol, respectively.

Heat of formation for the reaction at 30 °C:

= {[2 × (-64.62) +(-94.0518)]- [(-198.5) + (−26.4157)]/1000000} × 160.3787 = 0.2604 GCal

Heat of reaction for Reaction 1:

$[(\text{Enthalpy})_{FeO} + (\text{Enthalpy})_{CO2} - (\text{Enthalpy})_{CO} - (\text{Enthalpy})_{Fe2O3}] + (\text{Heat of formation})_1$

= [3.478 + 1.852-5.598-1.166]+[0.2604] = −1.1736 GCal

A.3. Model for coal consumption:

Expressions involved in the estimation of total heat requirement for conventional process are described below:

Hot utility requirement

Here, hot utility requirement, Q_{hu}, is determined by the pinch analysis [16]. The heat required for this utility is supplied by coal combustion.

Heat required by inlet air and bed material

Here, inlet air and iron ore are heated up to the reaction temperature in the preheating zone and then reduction takes place in the respective zone. The sensible heat achieved by air and iron ore is provided by combustion of coal and determined using following equations.

$$Q_a = M_a C_a (t_p - t_a) \tag{1}$$

$$Q_s = M_s C_s (t_p - t_a) \tag{2}$$

Heat demanded for the reduction process to continue

This heat is computed using heat balance of the process where heat of reaction is computed for reduction reactions, where Fe_2O_3 (Iron ore) is converted to Fe (sponge iron).

Heat lost through the kiln wall

Mainly heat lost through the kiln wall covers loss of heat through the shell, inlet and outlet hoods, post combustion chamber and the cooler inlet area. Hence, total heat lost is assumed as twice as that of the kiln [8,12].

$$Q_{loss} = (2\pi) D L h_r \tag{3}$$

Heat gained by coal

The coal is needed to be preheated to the reaction temperature and the sensible heat which is involved for this is shown below.

$$Q_c = M_c C_c (t_p - t_a) \tag{4}$$

Heat required for removing moisture of feed material

The average moisture contents are found in iron ore and coal are 2% and 13% by weight, respectively, in Indian conditions [17] and the amount of heat required to remove it is determined as:

$$Q_m = M_m \lambda \tag{5}$$

$$Q_m = (0.02 M_s + 0.13 M_c)\lambda \tag{6}$$

These amounts of heat are supplied by coal, thus:

$$Q_{hu} + Q_s + Q_a + Q_p + Q_{loss} + Q_c + Q_m = M_c \times NHV \times 0.433 \tag{7}$$

Based on heat balance, it is found that heat utilized in the process is only 43.3% of the total heat energy thus, term 0.433 is used in Eq. (8). Further, mass ratio of air to coal, practiced in actual plant is 3.8703. Thus,

$$M_a = (3.8703) M_c \tag{8}$$

Expressions shown through equations (1) and (4)–(6) and (8) are put in equation (7) and thus, equation (7) becomes

$$Q_{hu} + Q_s + 3.8703(M_c C_{pa} \Delta T_a) + Q_p + Q_{loss} + M_c C_{pc} \Delta T_c + ((0.02 M_s + 0.13 M_c) \times \lambda) = M_c \times NHV \times 0.433 \tag{9}$$

Now, air enters to the plant in three ways such as kiln air, in ABC and with slinger coal in mass fractions as 0.8504, 0.16107 and 0.03356, respectively. These are preheated up to different temperatures due to 600 °C, 200 °C, 200 °C respectively. Thus, different temperature approaches are applicable for these fractions such as ΔT_{a1}, ΔT_{a2}, and ΔT_{a3} are used for fractions 0.8504, 0.16107 and 0.03356, respectively. Considering these fractions and temperature approaches in Eq. (9) and rearranging the resultant equation the final expression is found as:

$$\begin{aligned} Q_{hu} + Q_s + Q_p + Q_{loss} + (0.02 \times M_s \times \lambda) = M_c [NHV \times 0.433 \\ -\{3.8703 \times ((0.8504 \times C_a \times (\Delta T_{a1})) \\ +(0.03356 \times C_a \times (\Delta T_{a2})) + (0.1611 \times C_a \times (\Delta T_{a3}))\} \\ -(0.13 \times 2256.9) - (C_c \times (T_d - T_{sc})] \end{aligned} \tag{10}$$

where, NHV = 22759.3 kJ/kg and $\lambda = 2256.9\ kJ/kg$

A.4. Determination of air requirement in the process

O_2 Required = Total O_2 consumed in combustion of [VM, C, H_2,CO]- Total O_2 present in coal

For complete process

In the rotary kiln total seven reactions are involved as reported in Section 2.

The calculation of O_2 consumed in the reaction 1

No. of moles of Fe_2O_3 is taken from Appendix A.2.

CO consumed = No. of Moles of CO × mol. wt. of CO = 160.3787 × 28 = 4490.602

CO2 produced = No. of Moles of CO_2 × mol. wt. of CO_2 = 160.3787 × 44 = 7056.661

Similarly, CO consumed in reaction 2 is computed as

CO consumed = No. of Moles of CO × mol. wt. of CO = 290.0186 × 28 = 8120.519

CO produced in reaction 3 is found as

CO produced = 2 × No. of Moles of CO × mol. wt. of CO = 2 × 450.3972 × 28 = 25222.24

CO available to react = (CO produced in reaction 3) - (CO consumed in reaction 1 and reaction 2)

= 12611.12

No. of moles of CO available for reaction = 12611.12/56 = 225.2

O_2 Consumed = 225.2 × 32 = 7206.4 kg

Similarly, O_2 Consumed for Reactions 4, 5, 6 and 7 are 5709.78, 1617.28, 481.332 and 481.332, respectively.

Total O_2 consumed [VM, C, H_2,CO] [Total O_2 consumed in the Rxn 3, 4,5,6,7]

= 7206.36 + 5709.78+1617.28 + 481.332+481.332 = 17132.33075

Total O_2 present in coal = 0.0659 × 13428 + 0.0667 × 7730 = 1400.4962

O2 required = [17132.33075−1400.4962] = 15731.834 kg/h

Air required (theoretical) = [15731.834/.21] = 74913.79788 kg/h

Air required (with 26%excess air) = [74913.79788 × 1.0931] = 81887.944 kg/h

A.5. Cost correlations

Cost of shell and tube heat exchanger = 1368000 + 34200 $(A)^{0.81}$

Cost of G-S HX = Factor size ratio × Standard cost of G-S heat exchangerWhere, Factor size ratio = (Load of finding G-S HX/load of standard G-S HX)

Cost of ceramic filter = Factor size ratio × Standard cost of ceramic filterWhere, Factor size ratio = (Respective flow rate of waste gas/capacity of standard filter)

Cost of duct = {(manufacturing cost + welding cost + installation cost)} of duct

Cost of FD fans = (No. of fans × cost of one ID fan)

A.5. . Cost calculation for Option-1

Considering straight line depreciation with 10 years as equipment life the total annual cost (TAC) is computed as Rs 12104.3 lakh/year. Also, total annual profit is computed considering revenue generated while exporting the power, expense of additional water requirement and maintenance costs of filter and power generation system and it is found as Rs 2029.9 lakh/year. Thus, the payback period of 4.17 years is predicted for Option-1 as shown in Table 5. The all cost is in Rs. lakh/year.

Amount of actual coal = 21.72 t/h

Amount of water consumption = 1242.7 t/h

Operating cost of coal = (2500 × 21.72 × 8000)/1000 = Rs 4344 lakh/year

Operating cost of water = (60 × 1242.7 × 8000)/100000 = Rs 5964.9 lakh/year.

Operating cost of ceramic filter = (90704971. × 0.9)/100000 = Rs 816.34 lakh/year

The term 90704971.17 comes by multiplying the standard operating cost of filter (112536000) with factor size ratio (0.806). Similarly the figure (14352000) is obtained.

Maintenance cost of turbine system = (14352000)/100000 = Rs 143.5 lakh/year

Total operating cost = {4344 + 5964.9+816.34 + 143.5} = Rs 11268.8 lakh/year

Total capital cost = Cost of steam boiler + Cost of ceramic filter = {((419796000 + 415731117.8)/10)/100000} = Rs 835.5 lakh/year

The term (419796000) comes by multiplying the standard cost of steam boiler with produced power generation.

Total Annual Cost (TAC) = Total operating cost + Total capital cost = 11268.8 + 835.5

= Rs 12104.3 lakh/year

Power generation = 8.97 MW [from Table 4]

Export = {((8.97 × 6.5 × 24 × 1000) × 365)/100000} = Rs 5107.518 lakh/year

Profit = {(556127.3506 × 365)/100000} = Rs 2029.9 lakh/year

The term (556127.35) is obtained by subtracting operating cost of filter and maintenance cost of steam boiler from (addition of the export cost and water).

Payback period = total capital cost/profit = {835527117.8/(556127.3506 × 30 × 12)}

= 4.17 year

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Nomenclature

C: specific heat, (J/kgK)
D: diameter of kiln, m
G: gas
h: heat transfer coefficient, kJ/h m^2
L: length of kiln, m
m: mass flow rate, kg/h
Q: Heat Load, kW
S: solid
T: temperature
t: tonne (=1000 kg)
NHV: net heating value, kJ/kg
CC: capital cost, Rupees
PI: Process Integration
T_{pa}: Primary air temperature
T_d: Kiln air
T_{sc}: Secondary air temperature
t_p: Reaction temperature
t_a: Ambient temperature

Subscript

a: air
c: coal
m: moisture
p: process
h_u: hot utility
s: ore, supply
t: target
r: radiation
loss: loss from kiln
i: inlet to kiln
pa: Primary air temperature
ΔP: friction (head or pressure loss) (inches water gauge/100 ft of duct)
q: air volume flow, (cfm: cubic feet per minute)
de: equivalent duct diameter (inches)
CC: annualized capital cost
OC: annual operating cost

Greek

λ: latent heat of vaporization, kJ/kg